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CESARAE: computationally efficient and statistical attack resistant audio encryption system

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Abstract

Audio piracy in the real-time transmission has intensified with digital replication and distribution, making unauthorized sharing effortless and widespread, leading to a significant hurdle in securing the original audio data. To counter this, computationally efficient and robust audio security measures like audio encryption play a pivotal role. In this paper, we suggest CESARAE, a computationally efficient, and robust against statistical attacks audio encryption system based on a new chaotic map called the Logistic Feedback Chaotic Map (LFCM). Firstly, we have developed and tested a new chaotic map LFCM, by adding a controlled feedback coefficient to a conventional logistic map. LFCM has been tested in terms of the Lyapunov exponent, bifurcation diagram, time series plot, phase trajectory, frequency assessment and NIST analysis and it has been found to have improved dynamic behavior and chaotic range making it applicable for real-time applications. Subsequently, LFCM has been used for the development of an audio encryption system, CESARAE. The proposed encryption system works in three stages. The original audio A_{org} to be encrypted is transformed into a grayscale image I_{audio} in the first stage. The auditory image is substituted in the subsequent stage, and a concurrent substitution and permutation procedure is carried out in the final step to obtain encrypted audio. CESARAE has been found to meet security goals, yielding average metrics of Correlation Coefficient CC = 0.0032, Number of Sample Change Rate NSCR = 99.62%, Unified Average Change Intensity UACI = 33.53%, Root Mean Square RMS(A_{org}) = 0.111, Crest Factor CF(A_{org}) = 13.7322, Root Mean Square RMS($Audio_{encrypt}$) = 0.3001, CF($Audio_{encrypt}$) = 5.3737, Signal-to-Noise Ratio SNR = 8.4035, Peak Signal-to-Noise Ratio PSNR (A_{org}, $Audio_{encrypt}$) = 4.0816, and Peak Signal-to-Noise Ratio PSNR (A_{org}, $Audio_{decrypt}$) = 59.3631,coupled with an expansive key space of $2^{300.207}$—far surpassing existing methods. Empirical outcomes underscore its robust security and satisfactory encryption/decryption speeds, rendering it a viable solution for secure communication and real-time data transfer.

Keywords Chaos · Security · Audio encryption · Robustness · Real-time systems

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1 Introduction

The rapid expansion of the digital realm and the widespread adoption of wireless audio communication technology have transformed our communication methods. Audio transmission over wireless channels is prone to various vulnerabilities as depicted in Fig. 1. More than 81.76 million records have been compromised over the last 12 months in 683 incidents (HIPAA). Thus, the increasing frequency and scale of data breaches emphasize the paramount necessity for comprehensive security strategies to defend against cyber threats and safeguard valuable information. While there are several security methods available, cryptography emerges as one of the most effective among cutting-edge techniques. It guarantees the confidentiality and integrity of audio data during its secure transmission making research on voice data encryption is of the utmost importance.

Numerous conventional encryption techniques, including the S-box algorithm [1], Data Encryption Standard (DES) [2, 3] and Advanced Encryption Standard (AES) [4, 5] are frequently employed in text encryption and produce effective encryption outcomes. Despite the fact that these algorithms use excellent encryption technology, some of them have flaws and do not provide foolproof security. The main weakness of AES (Advanced Encryption Standard) is its use of a fixed Substitution-box in its methodology, which diminishes its security and renders it susceptible to various combinatorial attacks. To guarantee the confidentiality, privacy, and authenticity of applications that rely on multimedia, it is considered essential to establish secure, fast, computationally efficient and dependable security measures.

However, when it comes to audio data, a lot of the encryption methods now in use have severe drawbacks. Due to their initial design for text or binary data, traditional techniques like AES and DES frequently find it difficult to handle the strong data correlation and real-time processing requirements of audio signals. These algorithms may also be computationally demanding, which can result in increased power consumption and slower processing times. These are significant disadvantages in situations with limited resources, such as embedded and mobile systems. This emphasizes the necessity of creating encryption methods that are both resilient to many types of assaults and tailored to the particular difficulties presented by audio data.

A crucial topic of research is the ability to fulfil the safety objectives of these trendy media consumption devices [6]. Entertainment platforms frequently use and heavily rely

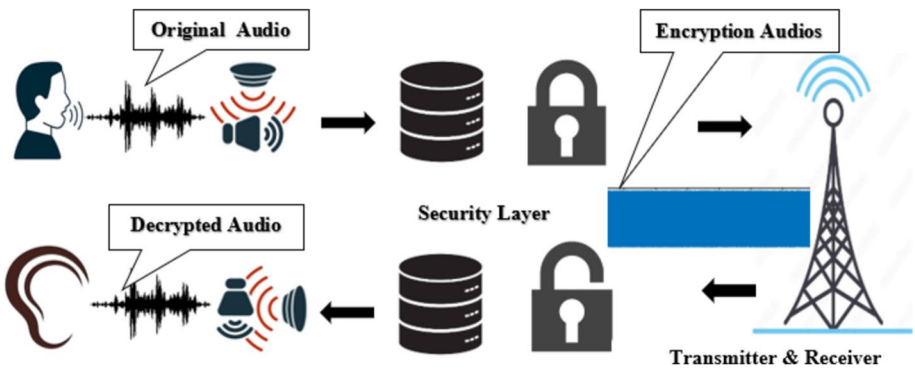


Fig. 1 Real-time scenario of secure audio transmission

on sound information applications [7, 8]. In many industries, including defence, medical science, telephone financial services, educational institutions, and secret audio conferencing, digital audio communications are becoming more and more common and prominent. The safety and endurance of those applications will be greatly impacted by audio information privacy concerns. In contrast to images, audio signals need instantaneous processing because they have a large magnitude, and have a significant relationship between data samples. Modern audio or voice applications also demand quick computation time and low power consumption.

Nonlinear dynamical systems, notably chaos, have made significant advances in recent years. The chaotic solutions have distinct characteristics such as hypersensitive dependency, ergodicity, and pseudo-randomness, which adds significant unpredictability to deterministic nonlinear systems. Chaos is most commonly used in multimedia encryption applications [9]. Chaos theory is currently identified as a factor of permutation and substitution in encryption [10]. The chaotic system behaves dynamically, furthermore, it is extremely reactive to control parameters and initial variables. It indicates that even a minor alteration in the initial parameters results in a significant transformation in the outcome. This factor increases the system's randomness. They are widely used in image encryption methods [11, 12]. Because of their superior cryptographic properties, chaos-oriented audio encryption methods are gaining popularity. These algorithms deal with redundant and large audio data, resulting in a high level of security.

A multi-chaotic mapping audio encryption system has been presented [13] with the following innovative features: dynamic shuffling and cryptography feedback are employed to achieve wide randomization along with extensive dispersion, respectively. Simultaneously, four distinct audio encryption algorithms which are self-adaptive randomization, various chaos-based maps, encrypted feedback, along with DNA encoding are merged into one system to increase security [14] proposes multilayer cryptosystems combining audio and speech signals for secure communication. Three encryption levels (fusion, substitution, and permutation) are considered, showing improved security compared to other techniques in evaluation metrics. In [15] the author has employed machine learning techniques and advanced feature selection methods to enhance DDoS attack detection in SDNs, demonstrating improved accuracy and efficiency. Authors in [16] have suggested another audio encryption technique based on the Substitution Permutation Network (SPN). Through Mobius transformation and Henon map, the algorithm achieved confusion and diffusion [17] proposes a secure audio encryption method using the chaotic Chebyshev map. Preprocessing and chaotic iteration scramble and diffuse the audio samples, ensuring lossless encryption. Security analysis confirms its suitability for secure voice transmission applications [18] demonstrated a unique audio encryption system based on block ciphering through DNA encryption and a logistic map. Channel scrambling is also used to prevent cryptanalysis.

To meet the needs for enhanced security in audio and device interactions, [19] developed Over Hear, a keystroke inference framework that uses Mel Frequency Cepstral Coefficients (MFCC) analysis and machine learning models to predict keystrokes from both acoustic and accelerometer data collected from headphones. To meet the needs of live low-profile acoustic applications in confined spaces, Researchers in [20] proposed a voice authentication method based on dynamic maps and DNA. The disadvantage of lightweight encryption is that they are susceptible to transmission noise. As a result, the scholars concentrated on creating robust schemes for applications utilizing multimedia. Robust ciphers, on the contrary, have a high computational burden and, thus are insufficient for platforms with limited computational capacity. Several noise-tolerant image

encryption techniques have recently used chaos [21] proposes a novel speech encryption algorithm using two chaotic maps; Henon and Gingerbread chaotic maps. The algorithm demonstrates effective randomness and achieves robust speech file encryption. Authors in [22] introduce a real-time chaos-based audio encryption method employing a novel chaotic map and a Pseudo-Random Bit generator. To meet the needs for efficient signal processing, [23] implemented an SSL algorithm using stochastic computing (SC). This SC-based system processes pulse-width-modulated signals, reducing area by up to 39% and power by 6%. A new cross-correlation design further cuts area and power usage by up to 64% and 82%, respectively [24] developed a branch-and-price (B&P) framework to address the issue of cognitive radio resource allocation with the Maximal Aggregation Range (MAR) constraint. This framework aims to efficiently manage spectrum allocation for secondary users (SUs) despite hardware limitations. In [25] the authors developed a novel audio encryption technique that is based on dynamic systems and DNA encryption. The method provides enough safety and has a high SNR (signal-to-noise ratio) [26] presents a novel audio encryption technique that is developed on Chen Memristor nonlinear system and FWHT (Fast Walsh Hadamar Transform). The results of the experiments and evaluations reveal that this technique is appropriate for many varieties of audio streams and can withstand many popular cryptographic assaults. When contrasted with prior acoustic encryption techniques, the algorithm's safety index is higher, and its performance is substantially increased.

According to the aforementioned evaluation of existing research, no sizable research is accessible for modern contemporary low-profile acoustic systems with noise-handling capability, particularly to solve safety concerns associated with wartime conversations in conflict zones. Furthermore, the chaos-based audio encryption techniques are mostly devoid of cryptanalysis, and insufficient safety assessments were carried out to ensure the stability of these cryptosystems against assaults. In this paper, we present a computationally efficient, and robust audio encryption system called CESARAE. The heart of CESARAE is newly proposed chaotic map. The following are major contributions of our proposed system:

- A new chaotic map LFCM, has been developed and tested.
- LFCM shows improved chaotic properties and can generate complicated pseudo-random patterns that are harder to predict than typical chaotic systems.
- LFCM has been used to develop CESARAE, a computationally efficient audio encryption system highly resistant to differential attacks.

The remaining portion of the paper is organized as follows. Section 2 describes the preliminaries. In Section 3 the proposed logistic feedback map is described. The audio encryption technique is presented in Section 4. Section 5 presents the experimental results and security analysis. Finally, the conclusion of the proposed scheme is drawn in Section 6.

2 Preliminaries

In this section, we provide the necessary background information and concepts to facilitate the understanding of the subsequent sections. The key theoretical foundations used in this work are as follows.

2.1 Logistic map

The logistic equation, which describes the variation of a single variable at discrete time intervals, defines the logistic map, a discrete-time dynamical system. It is expressed mathematically as

$$x_{n+1} = r \times x_n \times \left(1 - x_n\right) \tag{1}$$

where r is the control parameter and has a range [3.57, 4] and x_n is the initial value having range [0 1].

2.2 Hash function

A technique that can take a variable length input and produce an output of a fixed size is called a hash function. Its encryption and data integrity verification capabilities make it a popular choice. The Secure Hash Algorithm 512 bits, or SHA-512, is a member of the SHA-2 family of hash functions. Foremost than its predecessor, SHA-256, it generates a hash value that is 512 bits in length. Data authenticity is assured by SHA-512 hashing for the reason that the hash value it produces is substantially distinct for even slight alterations to the input. Apart from this, it is an ideal tool for audio safety due to its collision-avoidance features.

3 Proposed Logistic Feedback Chaotic Map (LFCM)

By incorporating various effective feedback state values and altering non-linear coefficients in a dynamic system, strengthened its intricate behaviour. This gives a wider parameter range and intricate patterns, making the dynamic system even more difficult to predict over time. The added stronger intricacy obtained from these modifications produces more dynamic and intricate behaviours, encouraging a more chaotic element of the dynamic system. Keeping all these factors in view, we have developed an effectual, competent, lightweight, highly sensitive, complex and resource constraint chaotic map LFCM that is a modified edition of the classical logistic map, the various steps constituents are:

Step 1 We have incorporated extra non-linear and complex coefficients M and R^2 in the classical foundational map. Subsequently, this enlarges the parameter spectrum, amplifies intricacy, and enhances sensitivity to initial controlling parameters in chaotic maps.
Step 2 Besides this, we have employed the two previous state values of the proposed map as feedback to the proposed LFCM and this feeding process will repeat every time for each iteration, allowing more accurate modelling of intricate behaviors and adaptive relationships.
Step 3 This conceptual methodological implication enhances nonlinearity, widens the parameter range, and raises overall complexity of LFCM. The mathematical representation of the proposed LFCM is as follows.

$$x_{n+1} = mod\left(\left(R^2 \times x_n \times \left(1 - M \times x_{n-1}\right)\right), 1\right) \tag{2}$$

where; R and M are the two control parameters. R has a range [0.3 ∞] and M has a range [0.1 ∞] moreover, x_n and x_{n-1} are the two initial values having a range [0 1].

The evaluation of the proposed map is done in terms of a bifurcation diagram, Lyapunov exponent, phase trajectories, NIST evaluation and time series plot which are discussed below in detail.

3.1 Bifurcation diagrams and frequency assessment plots

A bifurcation diagram depicts a dynamic system's behaviour as a control parameter is changed. It reveals the system's stability, chaos, and attractor structure. By examining the diagram, we can identify regions where the system exhibits various behaviors. In the case of the Logistic map and LFCM, the bifurcation diagrams displayed in Fig. 2 (a–e) shows that the Logistic map is chaotic for a specific range of the control parameter, while the LFCM map is chaotic over a wider range. The LFCM demonstrates chaotic behaviour across the entire parameter range, indicating an increased control parameter range compared to the Logistic map. Furthermore, we have assessed the power spectral density and frequency spectrum of the suggested chaotic map LFCM. The results attained, as illustrated in Fig. 2f & g, show that the proposed LFCM is intricate and tumultuous, as confirmed by power spectral density (PSD) and frequency spectrum assessment. A large, continuous frequency spectrum displays behaviour on multiple scales, and the 1/f PSD pattern authorizes the recursive pattern and extended linkages. Primary to dynamical systems theory, the PSD's significant variation and lack of prevailing frequencies affirms its chaotic nature by signifying its unpredictable dynamics and sensitive reliance on initial conditions.

3.2 Lyapunov Exponent (LE)

Lyapunov Exponent can detect whether the system shows chaotic behaviour. A positive LE shows that neighboring paths in the system diverge exponentially, indicating that the system is sensitive to initial conditions and has chaotic behaviour. A negative Lyapunov exponent suggests that neighboring trajectories converge, signifying stability and consistency. A Lyapunov exponent of 0 indicates neutral stability, in which paths do not diverge or converge but instead remain parallel. The LE comparison between the logistic map and the LFCM is displayed in Fig. 3.

As it is evident from the figure that the Logistic map shows a chaotic state only to the parameter range of 4, but LFCM has a positive Lyapunov exponent for the entire range of control parameters, that indicates the map exhibits chaotic behaviour much early then logistic map. Moreover, the LE value increases over time indicating an increase in the sensitivity.

3.3 Time series plot

The graphical depiction of the sequence of values produced by the map as it develops over subsequent iterations is known as a "time series plot". It offers a graphic representation of the dynamic behaviour and temporal development of the system under investigation. From

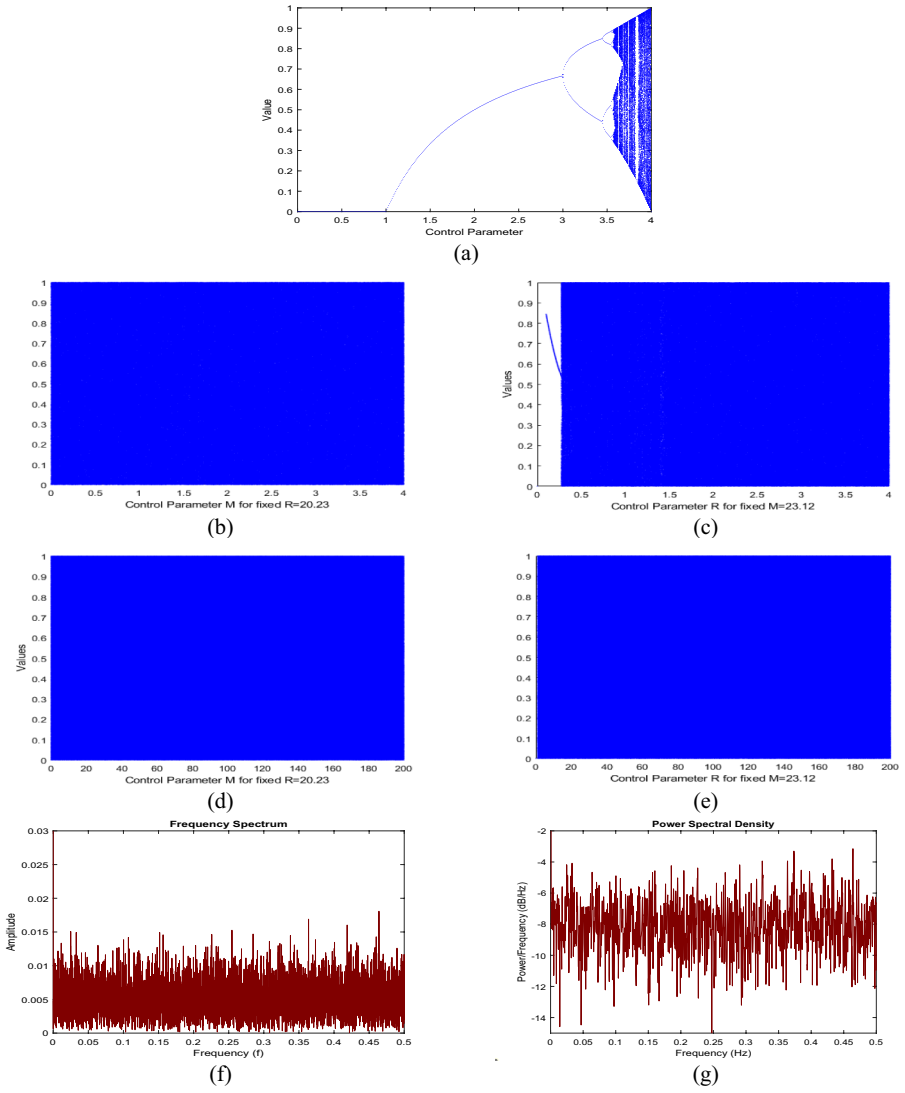


Fig. 2 Bifurcation diagram of (**a**) Logistic map (**b** & **e**) LFCM for the control parameter M at fixed R. (**c** & **d**) LFCM for the control parameter R at fixed M and (**f** & **g**) are the frequency assessment plots

Fig. 4, it is clear that the map has random behaviour and is highly sensitive to the initial conditions.

3.4 Two-dimensional and three-dimensional phase plot trajectory

A 2D phase diagram presents a visual depiction of the attractor structure and long-term behaviour of the system by displaying the link between two specified variables or system elements. It enables a meticulous knowledge of the progression and

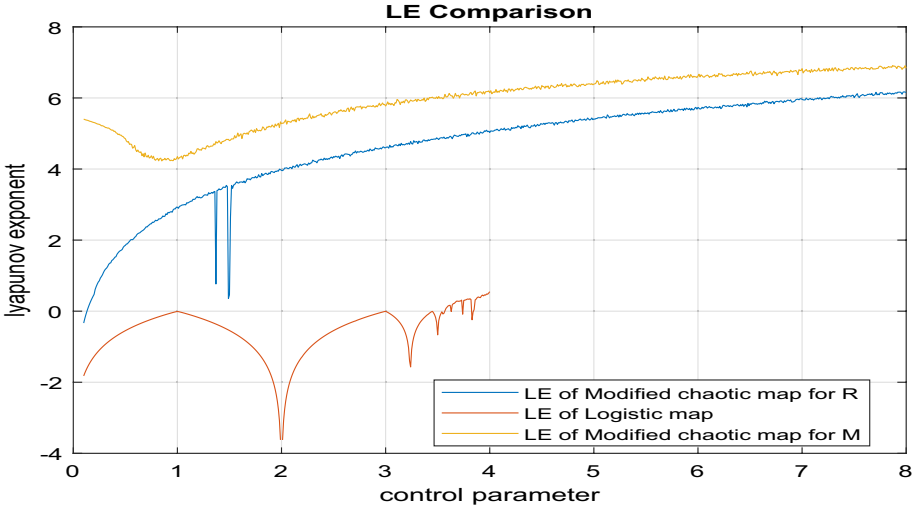


Fig. 3 Lyapunov exponent comparison

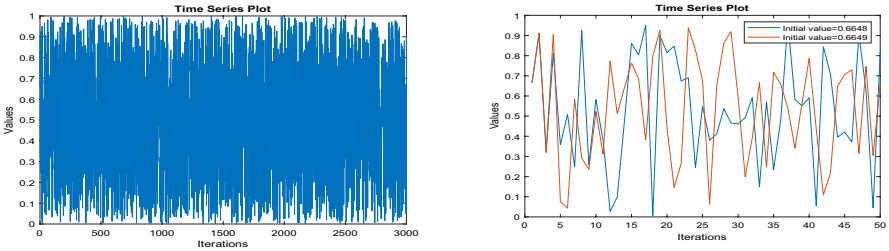
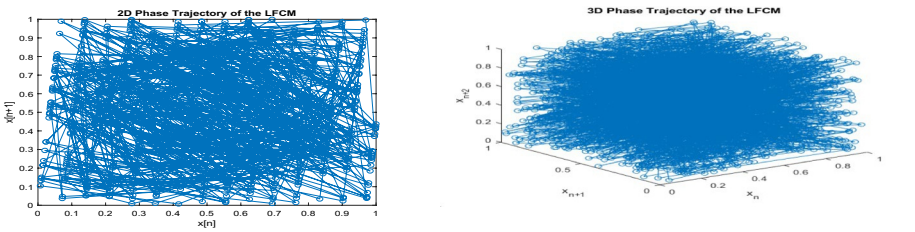


Fig. 4 Time series plot of (**a**) LFCM map (**b**) LFCM for two different initial values



(a) (b)

Fig. 5 (**a**) 2D and (**b**) 3D phase plot trajectory

Table 1 Analysis of NIST tests on Logistic map and LFCM

Test name	*P*-value Final Outcome		
	Logistic	LFCM	Result P > 0.01
Block Frequency Test	0.5681	0.9989	✓
Frequency Test	0.9005	0.5596	✓
DFT Test	0.8711	0.6380	✓
Longest Run Test	0.4746	0.6115	✓
Run Test	0.1849	0.8926	✓
Cumulative Test	0.9555	0.6778	✓
Binary Matrix Rank Test	0.2701	0.7012	✓
Non-overlapping Template Matching Test	0.9907	0.8932	✓
Overlapping Template Matching Test	0.2610	0.8073	✓
Maurer's Universal Statistical Test	0.9673	0.9021	✓
Linear Complexity Test	0.9856	0.8902	✓
Serial Test	0.3644	0.4371	✓
Approximate Entropy Test	0.0607	0.9417	✓
Random Excursion Test	0.2670	0.9955	✓
Random Excursion Variant Test	0.3965	0.4864	✓

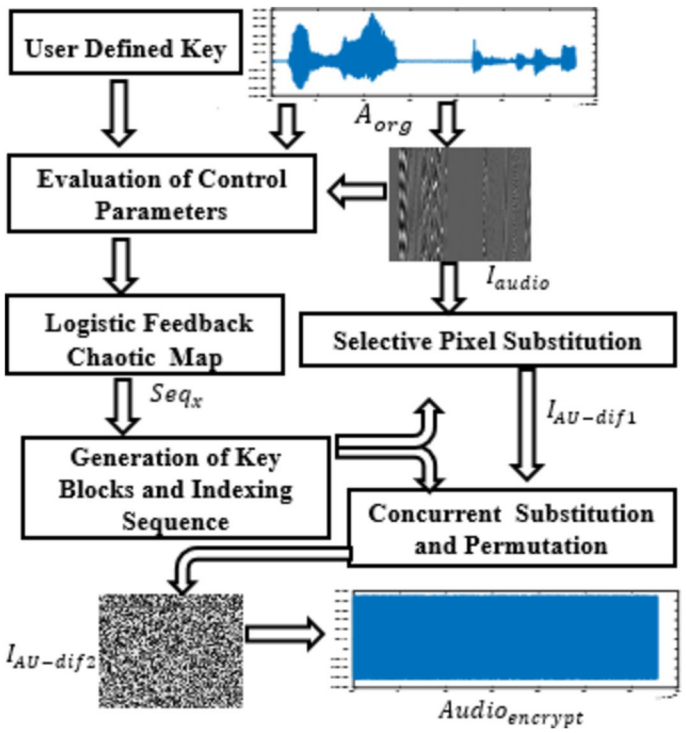


Fig. 6 Flowchart of encryption

collaboration of these variables through time, highlighting attractors, intricate patterns, and zones of stability, recurrence, and chaos. By incorporating an additional variable of the chaotic map, a 3-D phase diagram increases the 2-D phase diagram's visualization competencies. As the association between the three opted variables is persistently displayed, a more exhaustive insight into the dynamics of the proposed system is presented in Fig. 5 which depicts the 2-D and 3-D phase plots, where the number of iterations is taken as 3000.

The intricate dynamics and behaviour of the system are vividly represented in Fig. 5. The intricate patterns and non-linear correlations found in a chaotic system have been made obvious by the portrayal of the trajectory in Fig. 5.

3.5 NIST analysis

NIST analysis is used in encryption because it provides a standardized and rigorous evaluation of cryptographic algorithms. By following NIST guidelines, scholars can ensure their encryption methods are robust, secure, and widely accepted, fostering interoperability and trust among different systems and organizations. Table 1 represents the analysis of various NIST tests. The p-values for all of the NIST tests given in Table 1 are within the permissible range [0,1], suggesting that each test is successful.

4 The proposed encryption system (CESARAE)

This section presents the block diagram of CESARAE, the proposed audio encryption system. The architecture is built upon the substitution permutation network (SPN) structure and the primary three main components are audio image conversion, substitution and permutation. The flowchart of the cryptosystem is depicted in Fig. 6.

The key blocks produced by the LFCM are floating point numbers ranging [0, 1], which are not compatible with the encryption process directly. Thus, the formula for producing a random sequence is the prior processing of a chaotic signal. The productive numerical span of digital audio data is four decimal places, while the numerical range of the captured speech signal is [-1, 1]. The general concept is shown in Fig. 6 where a file of audio with size A is used as a source of input to the encryption scheme. The values of the samples in the original audio fell inside $[-1\ 1]$. To make the audio match the suggested design, a prior processing step is required. Resulting in a series of numbers around 0 and 255. The series is additionally divided into a range with a length of 65535 and then transformed into a matrix of size 256×256.

4.1 Audio signal pre-processing

Pre-processing of the audio signal is an important step in our methodology. We first acquire the audio and then separate out a one-channel portion for additional examination. The operations used to convert the audio signal into an image format are shown in Algorithm 1 given below.

Input: Original Audio A_{org}

Output: Audio-Transformed Image I_{audio}

Start Algorithm
Load the Audio file.
$min \leftarrow \min(Audio)$
$B \leftarrow Audio - min$
$C \leftarrow B/\max(B)$
$D \leftarrow C \times 255$
$I_{audio} \leftarrow reshape(D, [m \times n])$: Where m and n are the row and column of the transformed image.
End

Pre-processing of the original audio signal A_{org}, is a crucial initial segment in our cryptosystem that aimed at transforming the A_{org}, content, into a format suitable for cryptographic investigation and further processing. The various steps involved are as follows:

Step 1 The procedure begins with fetching the original audio file, subsequently, the minimum value of the audio signal A_{org}, $\min(A_{org})$, is evaluated to form a baseline.
Step 2 Thereafter, the audio signal is normalized by subtracting its minimal value, emerging in $B = A_{org} - \min(A_{org})$.
Step 3 To assure equiform intensity throughout the signal, B is then scaled to fit inside the range [0, 1] with $C = B/\max(B)$.
Step 4 This resulting signal C is enlarged to a limit appropriate for image interpretation, specifically [0, 255], via multiplication by 255, resulting $D = C \times 255$.
Step 5 Finally, D is transformed into an 8-bit unsigned integer layout, uint 8(D), and restructured into an image matrix I_{audio} with size $m \times n$ where m and n correspond to the rows and columns of the resultant image matrix.

This transformation procedure effectively translates the audio signal into an image format, set for subsequent cryptographic operations and detailed assessment within our research framework.

4.2 Generation of key blocks and indexing sequence

The core of our strategy is the application of the suggested LFCM algorithm to produce distinctive pseudo-random numbers. The important steps in the encryption process are the generation of key blocks and the indexing sequence. These elements form the basis of a safe encryption system that gives the audio data numerous levels of security. The operations used to generate key blocks and indexing sequences are shown in algorithm 2 given below.

Input: $x_0, x_1, I_{audio}, Str^{39}$ and C

Output: $Block_1, Block_2, Index_1$ and $Index_2$

Start Algorithm

$s_1 \leftarrow \sum I_{audio}(all)$

$s_2 \leftarrow (\sum C(all) \times 10^{-3})$

$s \leftarrow s_1 \times s_2$

$H_{V1} \leftarrow SHA_{512}(s)$

$H_{V2} \leftarrow SHA_{512}(Str^{39})$

$$R \leftarrow mod\left(\left(\left(\left(sum\left(H_{V1}(:)\right) + sum\left(H_{V2}(:)\right) \times S\right) \times 10^{-6}\right) + \max_original_audio\right), LFCM_{RANGE}\right)$$

$$M \leftarrow mod\left(\left(\left(\left(sum\left(H_{V1}(:)\right) \times S\right) \times 10^{-6}\right) \times 255\right), LFCM_{RANGE}\right)$$

$Seq_x \leftarrow Iterate_{2mn}\left(LFCM(x_0, x_1, R, M)\right)$

$Z_1 \leftarrow Seq_x(1\ to\ mn)$

$Z_2 \leftarrow Seq_x(mn + 1\ to\ 2mn)$

$$Block_1 \leftarrow reshape\left(mod\left(\left(floor(Z_1 \times 10^5)\right), 255\right), [m\ n]\right)$$

$$Block_2 \leftarrow reshape\left(mod\left(\left(floor(Z_2 \times 10^5)\right), 255\right), [m\ n]\right)$$

$Seq_1 \leftarrow Seq_x\left(2m\ to\ 2m + (m-1)\right)$

$Seq_2 \leftarrow Seq_x\left(6m\ to\ 6m + (m-1)\right)$

$Index_1 \leftarrow sort\ (Seq_1, ascend)$

$Index_2 \leftarrow sort\ (Seq_2, descend)$

End

4.3 Encryption process

The framework of the proposed encryption process is composed on the concepts of confusion and diffusion. This framework consists of two stages and the steps involved are as.

Step 1 In the first stage of the encryption process, depending on the parity of the row number the pixels belonging to the odd row are diffused using $Block_1$ on the other hand the pixels of the even row are diffused using $Block_2$ which means the pixel values are substituted selectively using either $Block_1$ or $Block_2$.
Step 2 In the second stage, a concurrent substitution and permutation operation is carried out on the resultant diffused image $I_{AU-dif1}$ to improve encryption through simultaneous pixel value changes using $Block_1$, $Index_1$ and $Index_2$ that results in $I_{AU-dif2}$.
Step 3 The encrypted audio is then created by converting the modified data $I_{AU-dif2}$ into a 1-D double array $Audio_{encrypt}$. By converting the audio data into a cryptic format and safeguarding it against unauthorized access, these procedures work together to secure it. The operations used in the encryption process are shown in algorithm 3 given below.

Input: $I_{audio}, Block_1, Block_2, Index_1$ and $Index_2$

Output: $Audio_{encrypt}$

Start Algorithm
For i =1:1:m
For j = 1:1: n
If i ← odd
$I_{AU-dif1}(i,j) \leftarrow bitxor\left(I_{audio}(i,j), Block_1(i,j)\right)$
Else
$I_{AU-dif1}(i,j) \leftarrow bitxor\left(I_{audio}(i,j), Block_2(i,j)\right)$
End
End
End
End
For i = 1:1:m
For j = 1:1: n
$I_{AU-dif2}\left(Index_1(i), Index_2(j)\right) \leftarrow$
$bitxor\left(I_{AU-dif1}(i,j), Block_1(Index_1(i), j)\right)$
End
End
$Audio_{encrypt} \leftarrow I_{AU-dif2}{}_{\text{Post-processing}}$
End

Successful decryption of encrypted audio is achieved by performing the above steps in reverse order.

5 Analyses of experimental simulation and safety

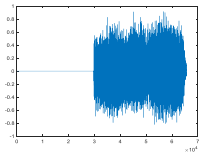
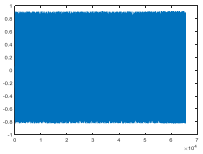
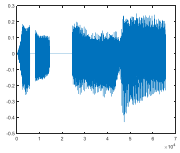
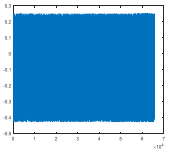
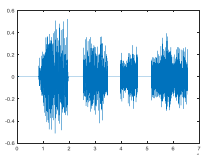
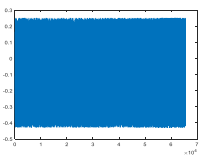
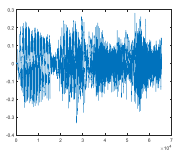
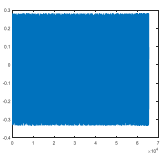
The experiment is being carried out in the MATLAB 2016a environment, with an Intel i5 CPU running at 1.90 GHz, Windows 10 (64-bit) and 4.0 GB of RAM. A number of tests were carried out to evaluate the performance of the suggested approach, four distinct audio files (MB) and amplitudes are designated as the evaluation targets for quantitative analysis of the experimental results to illustrate the algorithm's effectiveness. The spectrum diagram, statistical properties, key sensitivity anti-differential attack capability, RMS and peak factor, PSNR ratio, and encryption efficiency are examined and evaluated, respectively, to assess the security performance of the method. Audio in wav format is fed as a carrier input to the suggested technique in order to perform simulation evaluations.

5.1 Encryption effect

By encryption effect we mean how the cryptosystem transforms audio data into a non-readable format to ensure its confidentiality, integrity, and security from illegitimate access. Table 2 displays the experimental outcomes for various audio types following the encryption procedure. The semi-silent period Audio_1 (audio with low-level sounds and occasional silence), the non-silent period Audio_2 (continuous and audible sound without significant periods of silence), the intermittent silent period Audio_3 (audio with alternating periods of sound and silence), and Audio_4 are chosen as the test components of the technique to test its effects. The findings demonstrate that the distribution of the time series plot of the encrypted data following the encryption method is uniform and almost identical, demonstrating the algorithm's successful encryption.

The algorithm's impact after encryption and decryption is depicted in Fig. 7. The approach can successfully cipher the audio and deftly conceal it in the audio-encrypted plot, which in addition, accomplishes the desired encryption impact but also baffles the

Table 2 Original and encrypted audio data information

Audio_1 (Size 0.249 MB)		**Audio_2 (Size 0.249 MB)**	
Original Audio	Encrypted Audio	Original Audio	Encrypted Audio
			
Audio_3 (Size 0.249 MB)		**Audio_4 (Size 0.249 MB)**	
			

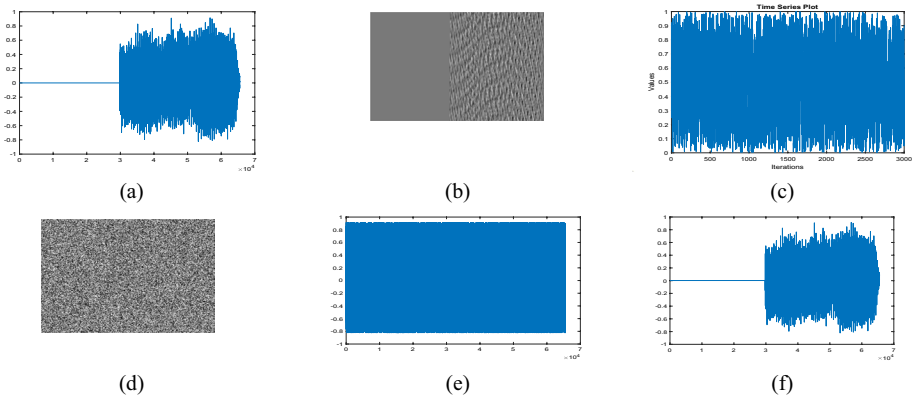


Fig. 7 Encryption and decryption of audio data: (**a**) original audio; (**b**) audio image; (**c**) time series plot of LFCM; (**d**) cipher audio image diagram; (**e**) cipher audio diagram; (**f**) decrypted audio diagram

attacker's audio and sight, greatly lowering the likelihood that audio data will be decoded after the interception.

5.2 Keyspace evaluation

Keyspace is the process of estimating the aggregate count of user defined keys that are utilized in an audio cryptographic system. The number of keys and the susceptibility of the keys may be used to evaluate the strength of the proposed cryptosystem against brute force attack. An efficient audio encryption technique should have a wide key space to offer a strong defence against brute force assaults. The range and combination of control parameters utilized in the suggested encryption strategy determine the key space. We intend to greatly increase the key space, ensuring a huge number of potential encryption keys, by carefully choosing and varying these characteristics. By increasing resistance to both common and complex attacks, this strategy will make illegal decryption almost impossible within a reasonable amount of time. We examine the discrete values of each control parameter and combine the possible values of all the parameters to determine the precise size of the key space. The initial parameters x_1 and x_2 can be represented by 5 and 4 decimal points respectively. Thus, the total count of possible combinations for x_1 and x_2 are 10^5 and 10^4 respectively. Also, a string with a length of 39 letters is used where each letter in a string

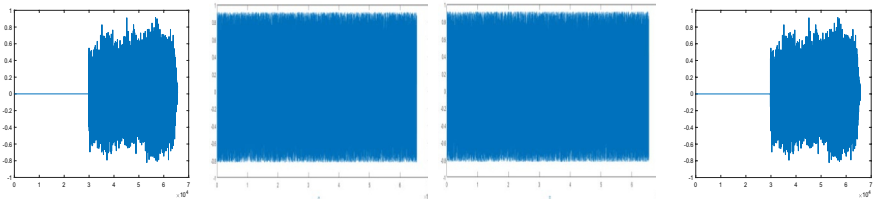


Fig. 8 Key sensitivity test: (**a**) Original audio file; (**b**) Alter key 1; (**c**) Alter key2; (**d**) Correct Key decryption

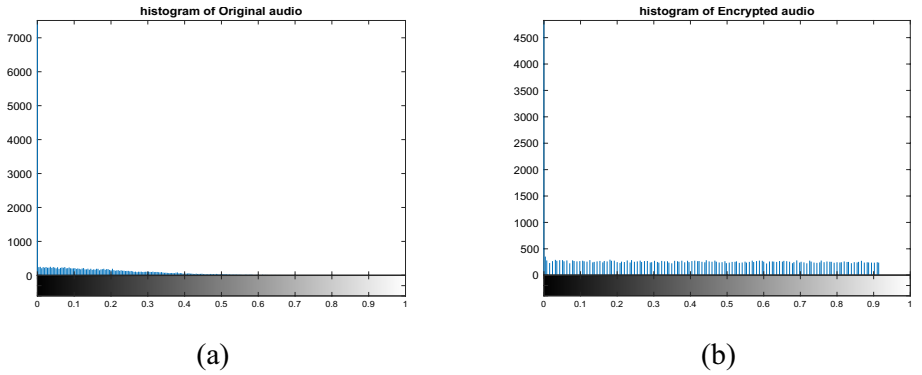


Fig. 9 Histogram: (**a**) original audio; (**b**) encrypted audio

can take 122 different values and hence, the total number of possible combinations for a string is 122^{39}.

$$Keylength = \log_2\left(\left(10^5\right) + \left(10^4\right) + \left(122^{39}\right)\right) = 300.207 bits \tag{3}$$

$$Keyspace = 2^{300.207} \tag{4}$$

A key space of $2^{300.207}$ offers a vast number of potential encryption keys, which offers incomparable cryptographic security. Because of the size of the key space, brute-force assaults are computationally impractical, providing strong security against unwanted access to private information. Because of the enormous size of the key space, which ensures an outstanding level of cryptographic strength and makes the encryption algorithm very resistant to contemporary computing and cryptographic advancements, data confidentiality and integrity in crucial applications are maintained.

Table 3 Autocorrelation of original, encrypted, and decrypted audio signals

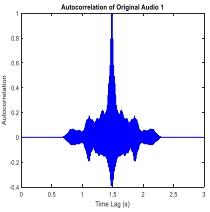
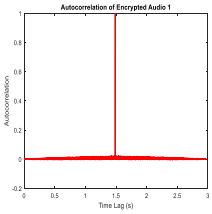
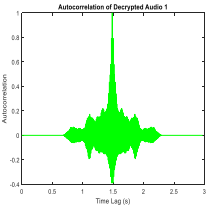
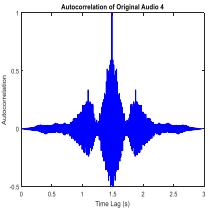
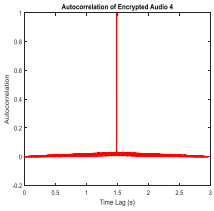
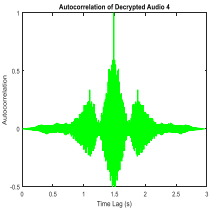
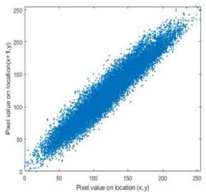
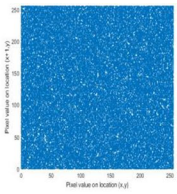
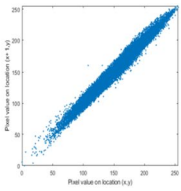
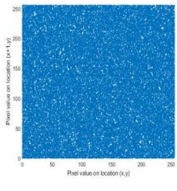
File_name	Original Audio	Encrypted Audio	Decrypted Audio
Audio_1			
Audio_4			

Table 4 Distribution of adjacent pixels of the original and encrypted audio image

Audio_1		Audio_2	
Original Audio	Encrypted Audio	Original Audio	Encrypted Audio
			

5.3 Key sensitivity

Examining a difference among two encrypted audio files produced by encrypting the exact same audio with slightly different keys is known as key sensitivity analysis. This investigation assesses how minor key adjustments affect the encrypted audio and reveals insight into how sensitive the encryption method is to key changes. An audio encryption technique must be highly sensitive to even the smallest changes in its keys to be declared robust. This suggests that a slight change in the key value must produce entirely different encrypted audio. Even small changes to the initial keys of the encryption process will result in improper decoding. Figure 8 shows the experimental results of decryption using faulty and accurate keys, emphasizing the critical importance of key correctness. The experimental results show that even minor changes in the algorithm's key can cause the decrypted audio to be inaccurately representative of the original audio input. Instead, it displays major discrepancies from decryption with the right key and a lot of audio noise. The algorithm is extremely secure due to its great sensitivity to the key and higher resilience to brute force assaults.

5.4 Histogram

Audio histogram evaluation involves examining the dissemination of audio signal amplitude values to ascertain underlying patterns that gives the insight about the robustness of the cryptosystem. It is the reliable way to evaluate the quality of encrypted audio transmissions. An efficient encryption method should yield a randomized noise with a nearly flat distribution of samples from the source audio file. By making the cyphered audio statistically similar to random noise, this property enables the encryption system to effectively face off statistical assaults. Histogram of the original audio is shown in Fig. 9.

The encryption procedure has successfully concealed any observable patterns or regularities that may have been present in the original audio, as indicated by a uniform histogram in the encrypted audio.

5.5 Correlational analysis

The statistical tool of correlation analysis is used to assess how well an encryption method defends against various statistical assaults. It measures how closely similar

parts of the original and the encrypted audio data files are related. Effective encryption ensures that there is less connection between the original and encrypted audio segments by converting the audio file into a randomized, chaotic signal with a minimal correlation coefficient.

5.5.1 Autocorrelation

It is the measurement of the cross-correlation of the signal with its own values. The results of autocorrelation analysis performed on various audio signals are shown in Table 3. The given below mathematical equation is used to compute the autocorrelation.

$$r_{AA}(\lambda) \triangleq E\left[\bar{A(i)}\, A(i+\lambda)\right] \tag{5}$$

here, λ denotes the coefficient of delay.

The autocorrelation of several audio signals is depicted in Table 3. The four encrypted signals' autocorrelation patterns all show the same noise characteristics. Additionally, there is a strong encryption effect and the possibility to perform lossless reconstruction since the autocorrelation plots of the source and the decrypted audio closely match their waveforms. This autocorrelation graphic consistency shows how well the technique protects the audio's integrity during encryption and decryption operations.

5.5.2 Correlation coefficient analysis

One typical method to assess the efficacy of encryption techniques is to compute the Pearson correlation coefficients between consecutive samples (horizontally) prior to and following audio encryption. A sound encryption system interferes with the correlation between neighbouring audio impulses, which is often highly correlated in the original audio. Creating low-correlation random noise from audio signals helps achieve the required encryption result. The following formula is used to calculate the correlation coefficient.

$$r_{ab} = \frac{cov(a,b)}{\sqrt{D(a)}\sqrt{D(b)}} \tag{6}$$

$$cov(a,b) = \frac{1}{N}\sum_{n=1}^{N}\left(a_n - E(a)\right)\left(b_n - E(b)\right) \tag{7}$$

$$D(a) = \frac{1}{N}\sum_{n=1}^{N}\left(a_n - E(a)\right)^2 \tag{8}$$

$$E(a) = \frac{1}{N}\sum_{n=1}^{N} a_n \tag{9}$$

The efficacy of the algorithm is assessed by randomly choosing 2000 pairs of neighbouring audio image pixels from a logarithm of the selected audio image pixels, N,

where a and b indicate two consecutive audio image pixels. The findings are shown in Table 4, which shows the correlation patterns between the encrypted and the original audio. This approach successfully reduces the correlation among the consecutive data points of encrypted audio, achieves an effective encryption result, and has great security, as can be observed from Table 4 that the correlation pattern of the encrypted audio is uniform and insignificant.

Table 5 displays the correlation coefficients between the encrypted audio and the original audio. Correlation represents the measure of their linear relationship and similarity. The ideal audio correlation coefficient in documents [1, 26] is much higher than the correlation coefficient found for ciphered audio in this research, showing that the encryption technique may efficiently minimize the correlation and thus indicate that the audio data is successfully dispersed.

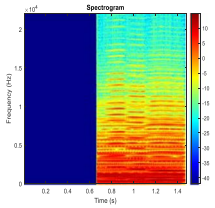
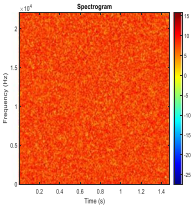
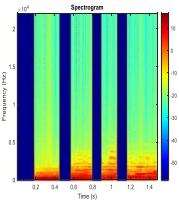
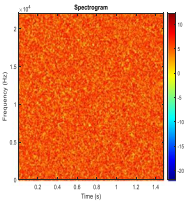
5.6 Spectral analysis

A spectrogram is a graphic depiction of the frequency spectrum that changes over time in an audio recording. With the Fourier transform, it is obtained. The two geometric aspects of frequency and time are used to depict it. The spectrum for each of the audio samples is computed using the Fourier transform after the audio samples have been separated into chunks. The decibel of audio transmissions at various frequencies and times are represented by distinct colors in the spectrum. Low decibels are represented by the dark blue region, whereas high decibels are represented by the dark red area. Table 6 displays the spectrograms of the original and the corresponding encrypted audio files. The distribution features of elevated decibel levels are compatible with the intensity plot of the original audio, as shown in Table 6. The frequency profile of encrypted audio is uniformly spaced and all of the frequencies are elevated decibels, indicating that the audio signal was efficiently dispersed and encrypted audio is quite high decibel sound without any distinguishing features.

Table 5 Correlation coefficients

Audio _1	Audio _2	Audio _3	Audio _4	Ref. [1]	Ref. [26]	Ref. [34]
-0.0020	-0.0055	-0.0042	0.0013	0.0031	0.0054	0.0088

Table 6 Spectrum plot

Audio_1		**Audio_3**	
Original Audio	Encrypted Audio	Original Audio	Encrypted Audio
			

5.7 Differential attack analysis

Differential attack analysis involves investigating how slight modifications in an input audio affect the cipher output, facilitating adversaries to observe patterns and possibly reveal susceptibilities in the proposed cryptosystem, compromising security. Differential attacks take advantage of variances in output values brought on by minuscule adjustments to input values. The goal of differential attack analysis is to determine how vulnerable cryptographic algorithms are to such attacks, spot any possible weak spots, and create efficient defences to strengthen their resistance to them. The encryption technique must be extremely sensitive to plaintext data, i.e., altering audio signal data while using the same starting key should produce different ciphertext sounds. Typically, measures like the NSCR (Number of Sample Change Rate) and UACI (Unified Average Change Intensity) are used to gauge how sensitive data is. The following is its definition.

$$NSCR = \sum_{n} \frac{D_n}{A_i} \times 100\% \tag{10}$$

$$UACI = \frac{1}{A_i}\left[\frac{\sum_n a_n - a_n\prime}{2^P - 1}\right], D_n = \begin{cases} 1, a_n \neq a_n\prime \\ 0, a_n = a_n\prime \end{cases} \tag{11}$$

Theoretically, the uniform average change intensity and sample number change rate should be close to 100% and 33.3333%, respectively [27]. With the use of an audio encryption scheme, two audios with a slight variation are converted into two corresponding ciphertext audio. Table 7 displays the average UACI and NSCR values for these four separate ciphertext audio samples after a minute alteration in the input audio. As compared to the previously existing state-of-the-are techniques, the NSCR and UACI in Table 7 are nearer to the ideal value overall, demonstrating the algorithm's high sensitivity to unencrypted audio and great immunity to differential assault.

5.8 RMS (root mean square) and CF (crest factor)

The average audio amplitude level source determines the value of the root mean square (RMS). When the input signal has a mean of zero, it is identical to the standard deviation and is calculated as follows:

$$RMS = \sqrt{\frac{1}{A_i}\sum_{n=1}^{A_i} |X_n|^2} \tag{12}$$

The peak-to-average ratio of a signal is quantified by the crest factor (CF). It offers important details regarding the signal's dynamic range or the range between the highest and lowest signal levels. CF is calculated as follows:

Table 7 NSCR and UACI values

File_name	Audio_1	Audio_2	Audio_3	Audio_4	Ref [1]	Ref [26]	Ref [35]
NSCR	0.9961	0.9961	0.9964	0.9960	0.9953	0.9992	0.9961
UACI	0.3350	0.3357	0.3351	0.3355	0.2579	0.3281	0.3340

$$CF = 20\log_{10}\frac{\left|X_{peak}\right|}{X_{rms}} \tag{13}$$

The RMS and CF (dB) values for different types of original and encrypted audios are displayed in Table 8. It is obvious from the Table that the CF values of encrypted audio files are less therefore indicating that the encrypted audio is uniformly dispersed which ensures good security. The outcomes as displayed in Table 8 depict the absence of a statistical correlation between the original audio and its associated encrypted audio.

5.9 Evaluation of parameters using custom audio dataset

By utilizing the unique collection of audio samples retrieved from human sources, we exhaustively assessed our audio encryption and decryption scheme to assess its robustness and efficacy. We have determined various significant properties, such as crest factor, signal-to-noise ratio, root mean square, peak signal-to-noise ratio, correlation coefficient and perceptual audio quality parameters. We extensively estimated our proposed cryptosystem's proficiency in real-time settings by revealing it to this audio dataset. These examined result findings depict how effectively the proposed approach protects audio quality while averting real-world attacks and defects. After these confirmation procedures, we are confident in the capability of our proposed cryptosystem. The results obtained from the custom audio dataset are displayed in Table 9.

We have employed a subjective hearing test with self-recorded personal audio to assess the sufficiency and quality precision of our suggested audio security system. The auditory accuracy, sensorial quality, and understandability of both the encrypted and decrypted audio were the main focus of this estimation. Subjective evaluations were approved by test respondents, and their replies were subsequently collected and estimated. We are pleased as the results of the subjective listening estimation came out to be positive. From Table 10, we can conclude that our audio cryptosystem works as expected, protecting data without compromising on its quality.

Table 8 RMS and CF

File_name	Audio_1	Audio_2	Audio_3	Audio _4	Ref [1]	Ref [2])	Ref [37]
RMS (A_{org})	0.1805	0.0769	0.1085	0.0812	0.0034	0.1389	-
RMS ($Audio_{encrypt}$)	0.5064	0.2155	0.2988	0.1799	-	0.5701	0.5772
CF (A_{org})	14.0981	14.9191	13.6523	12.2595	49.3	16.4115	-
CF ($Audio_{encrypt}$)	5.1389	5.9674	4.8581	5.5305	-	3.4499	4.9909

Table 9 Evaluation of parameters using custom audio dataset

CC	NSCR	UACI	RMS (A_{org})	RMS ($Audio_{encrypt}$)	CF (A_{org})	CF ($Audio_{encrypt}$)	SNR (dB)	PSNR (org, enc)	PSNR (Org, Dec
0.0036	0.9961	0.3345	0.1925	0.4818	14.3730	4.7895	-7.8007	5.3818	54.8556

Table 10 Subjective listening test feedback

Subject	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5
Result	Pass	Pass	Pass	Pass	Pass

5.10 Perceptional analysis

To assess the data quality, signal-to-noise ratio (SNR) and peak signal-to-noise ratio (PSNR) are frequently used metrics. SNR quantifies the amount of noise in an encrypted signal. Cryptography experts aim to elevate the noise level to effectively encrypt signal content. A lower PSNR value is needed for the encrypted file since it indicates the file has a high amount of noise and potent defence against assault. These definitions apply to PSNR and SNR between two separate audio sources.

$$SNR = 10 \times \log_{10} \frac{\sum_{n=1}^{i} X_n{}^2}{\sum_{n=1}^{i} \left(X_n - A_n\right)^2} \tag{14}$$

$$PSNR = 10 \times \log_{10} \left(\frac{\max^2}{mse} \right) \tag{15}$$

where X_n and A_n represents the original and encrypted audio respectively.

The obtained SNR values are all negative, indicating that the encryption process effectively obliterated the clear signal from plain audio files, making the encrypted files exceedingly noisy. The obtained PSNR values in Table 11 are all low positive values, which indicates that the cyphered audio files include a significant amount of noise. The PSNR between the original and decrypted audio is given in Table 12, which demonstrates that the proposed encryption technique is reversible and there is successful retrieval of the original audio from the encrypted file as the calculated value between the two is quite high.

5.11 Evaluation of asymptotic computational efficiency

In the proposed cryptosystem the audio input, consisting of N samples, undergoes an encryption process, where $N = P_{Row} \times P_{Col}$ in which P_{Row} and P_{Col} are any two natural

Table 11 Values of SNR and PSNR

File_name	Audio_1	Audio_2	Audio_3	Audio_4	Ref. [18]	Ref. [26]	Ref. [35]	Ref. [36]
SNR (dB)	-8.9593	-8.9517	-8.7942	-6.9091	-28.1400	-18.8850	−38.2807	-
PSNR (dB)	4.1860	4.0004	4.0433	4.0979	4.3100	4.8659	5.1042	4.7965

Table 12 PSNR between original and decrypted audio

File_name	Audio_1	Audio_2	Audio_3	Audio_4
PSNR (dB)	53.5033	61.1882	59.6197	63.1414

Table 13 Asymptotic complexity comparison of the proposed cryptosystem

Scheme	Ref [28]	Ref. [29]	Ref. [30]	Proposed
Big O	$O(N^2)$	$O(N^2)$	$O(N^3)$	$O(2N)$

numbers. There arise two cases, in the initial case $P_{Row} \neq P_{Col}$ we get non-square image matrix and second one is where $P_{Row} = P_{Col}$ where we obtain square image matrix, in both scenarios the cryptosystem is valid and the audio samples $N = P_{Row} \times P_{Col} = P^2$ where P^2 represents the total number of samples needed to obtain an accurate audio image. Our proposed cryptosystem's computational analysis is divided into three sections: audio pre-processing where entire audio samples are processes resulting with complexity $O(P^2)$, key blocks and indexing sequences evaluation has complexity equal to $O(2P^2)$ as in this we iterate the proposed chaotic map $2P^2$ times then from this chaotic sequence we further process it to obtain the required blocks and sequences as indicated in the Algorithm 2, and the proposed encryption process has complexity $O(P^2)$ as depicted in Algorithm 3, as all the samples are processed to obtain the required cipher audio. Consequently, the overall asymptotic complexity is logically $O(2P^2)$, it is effectively $O(2N)$ in comparison with existing methods, which typically consider N samples without specifying the quantitative sum. The results illustrated in Table 13 depict that the proposed cryptosystem is computationally efficient compared to existing standard methods.

5.12 Encryption speed and time estimation

Encryption time is essential for estimating the efficiency of the cryptosystem, especially for real-time processing. In this work, we have evaluated both encryption time and speed across various audio file sizes, revealing a linear association between file size and encryption time. This linear scaling is valuable as it allows accurate predictions of encryption time for any given file size, which is crucial in constraint applications. By measuring speed in kilobytes per second, we provided a clear understanding of the cryptosystem's computational efficiency. The results displayed in Table 14 demonstrate that our algorithm performs competitively in terms of encryption speed and scalability. This comparative analysis underscores the system's capability to efficiently secure different data sizes in a fraction of a second, making it suitable for real-time applications.

Table 14 Evaluation of encryption speed and time of the proposed cryptosystem

Size *Kb*	Speed $\left(\frac{Kb}{Sec}\right)$ Proposed	Speed $\left(\frac{Kb}{Sec}\right)$ Ref. [31]	Size *Kb*	Execution time(s) Proposed	Execution time(s) Ref. [32]
24.8	0.00019	0.0687	307	0.044	0.612
31.6	0.00037	0.0803	345	0.051	0.832
42.1	0.00040	0.1134	434	0.075	0.987
43	0.00045	0.1129	530	0.100	1.230
87.1	0.00103	0.2340	783	0.175	1.456
102	0.00142	0.2745	24000	2.990	4.653
822	0.01459	1.0331	26000	3.240	5.123
1,254	0.01798	0.6475	32000	3.990	6.341

Table 15 Chosen attack evaluation

$Exper_I$	$Exper_{I1}$	$Exper_{I2}$	$Exper_{I3}$	$Exper_{I4}$	$Exper_{I5}$
IND-CPA	22.00 %	58.00 %	48.50 %	15.00 %	35.00 %
IND-CCA	18.33 %	22.57 %	50.33 %	24.73 %	20.33 %

Table 16 Size scalability analysis of the proposed cryptosystem

Size of Audio(Mb)	NSCR	UACI	CC	RMS ($Audio_{encrypt}$)	CF ($Audio_{encrypt}$)	SNR (dB)	PSNR ($org, encrypt$)	PSNR (Org, Dec)
0.062	0.9961	0.3354	-0.0028	0.2159	5.9673	-8.9517	4.1261	61.1882
0.249	0.9961	0.3354	-0.0020	0.2155	5.9674	-8.9516	4.1256	61.1885
0.99	0.9962	0.3355	-0.0020	0.2151	5.9661	-7.9510	4.1121	62.2316
3.99	0.9964	0.3357	-0.0013	0.1899	5.9660	-6.9091	4.0003	63.1231

5.13 Security analysis against chosen attacks

We have conducted a thorough security analysis of our proposed audio cryptosystem, addressing both IND-CPA and IND-CCA attacks as discussed in [33]. Adversaries attempted to guess encrypted plaintexts and corresponding ciphertexts but consistently failed, with success rates below 50% across five iterations and fifty random choices per iteration. This demonstrates the cryptosystem's robust security, as adversaries couldn't reliably discern between distinct ciphertexts, as detailed in Table 15.

5.14 Size scalability analysis

We have evaluated all the parameters for various audio with variable size, as displayed in Table 16. The obtained outcomes signify that there is no compromise on the security and can effectively maintain its effectiveness as the workload or data size increases.

The evaluated parameters give a robust evaluation of the suggested cryptosystem's performance across critical aspects. Metrics notably the RMS, CC, CF, UACI, and NSCR, demonstrate its adaptability to data unsteadiness and noise, crucial for sustaining resilience under Inauspicious settings. PSNR and SNR evaluate its efficacy in retaining data confidentiality and validity, important for secure transmission. Computational overhead evaluation with asymptotic assessment provides insight into the real-time transmission capacity as supporting applications require speedy encryption and decryption. Keyspace and key sensitivity estimate further approve its security strength, analyzing resistance to key compromise and affirming sufficient key space to withstand cryptographic attacks effectively.

6 Conclusion

In the modern digital horizon, the expansion of audio security is a direct effect of effortless duplication and extensive distribution. To resolve this challenge, the adoption of robust security primitives assumes a predominant role. This paper presents an effectual audio cryptosystem CESARAE based on a newly suggested logistic feedback chaotic map (LFCM) holding complex dynamic behavior. The hands-on execution of this LFCM emphasizes the

algorithm's competence. Elevating the robustness of the proposed technique, the incorporation of a chaotic map is leveraged. The proposed encryption process comprises three fundamental phases. Initially, the original audio A_{org} undergoes transformation into a grayscale image I_{audio}. Subsequently, the auditory image is subjected to a substitution process, followed by a concurrent procedure of substitution and permutation in the final stage. Upon completing these operations, the outcome is an encrypted audio image, which is then converted back into the audio format $Audio_{encrypt}$ that results in the desired encrypted audio file. The suggested encryption scheme offers an advanced standard of security in contrast to the existing state-of-the-art methods, ensuring the secure communication of real-time audio in various applications. In future studies, our aim is to examine possible advancements to the encryption system, such as adding additional cryptographic primitives and evaluating how well it performs under various real-world attack scenarios. Additionally, expanding the evaluation to a larger and more varied dataset would give stronger insights into the robustness and suitability of the algorithm in real-world situations.

Data Availability Data cannot be made available due to the Intuition policy.

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Ethical Approval All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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